

Eigenvalues and Eigenvectors

$A \leadsto$ a $n \times n$ matrix
 $x \leadsto$ non-zero vectors

Eigenvalues and eigenvectors
 are a new way to see into
 the heart of a matrix.

So, if Ax are parallel to x , then they are eigenvectors.

$\hookrightarrow$ So we want the vectors that don't change the direction of x .

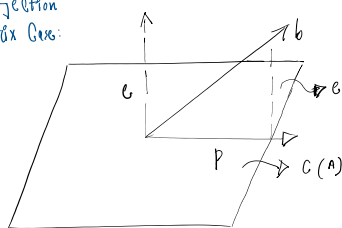
$\hookrightarrow Ax = \lambda x \leadsto$

$\lambda \rightarrow$ Eigenvalues
 $\lambda x \rightarrow$ Eigenvectors

⊗ If A is singular (that does not have an inverse)
 then at least one λ would be zero.

⊗ 0 vector is always an eigenvector for any matrix.

Projection
Matrix Geo:



⊗ For projection matrix P , the column space
 projects to itself ($Px = x$).

So, vectors in column space has eigenvectors with $\lambda = 1$.

⊗ The nullspace of P projects to zero ($Px = 0x$)

So, vectors $\perp$ to column space eigenvectors with $\lambda = 0$

Example:

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

$\hookrightarrow$ After solving this using $Ax = \lambda x$, we will get $\lambda = 0, 1$
 (We will see how has this been done later on) ⊗

Now, what are the properties of P here?

$\hookrightarrow$ ⊗ Markov Matrix $\rightarrow$ A matrix where each column sums to 1 is a Markov matrix.
 Here, both columns sum to 1: $0.5 + 0.5 = 1$. This means P preserves probability distributions.

$\hookrightarrow$ ⊗ Singular matrix $\rightarrow$ P is singular because it has zero determinant. (One of its eigenvalues is 0)

$\hookrightarrow$ ⊗ Symmetric Matrix $\rightarrow$ Since $P^T = P$, it is symmetric, which ensures orthogonal eigenvectors.

Some key-points: $\Rightarrow$

Proof:

- A projection matrix **only has eigenvalues 0 and 1**.
- **Eigenvectors for $\lambda = 1$ remain unchanged** (they form the subspace being projected onto).
- **Eigenvectors for $\lambda = 0$ are destroyed** (they are in the null space). ⊗
- The projection matrix is **singular, symmetric, and a Markov matrix**.

- Projection removes components along the null space while preserving the subspace.

Let us come to the main part now?

Now how to solve $Ax = \lambda x$? Before that let us go through the solvability conditions

① A must be square. Let's say A is $m \times n$. If $A \begin{matrix} x \\ x \\ x \end{matrix} = \frac{\lambda x}{\lambda x} \rightarrow A$ takes 2D vector, makes it 3D vector. Vector is now out of the plane. As direction is changed, no eigenvectors are possible.

$\hookrightarrow$ Eigenvalues and Eigenvectors are only defined for square matrices because they describe how a transformation scales vectors within the same space.

$\hookrightarrow$ But does all square matrix have eigenvectors and eigenvalues? No! But why? Will explain later on. *

② Only symmetric matrices would have a set of real, orthonormal eigenvectors.

$\hookrightarrow$ This will be proved later on. *

$\hookrightarrow$ For non-symmetric matrices may still have eigenvectors, but:

$\hookrightarrow$ Some of them might be complex.

$$\hookrightarrow \underbrace{(A - \lambda I)}_{A - \lambda I} x = 0$$

$A - \lambda I \rightarrow$ must be singular. Otherwise the only x here would be 0

$\hookrightarrow \det(A - \lambda I) = 0 \rightarrow$ Characteristics equation / eigenvalue equation
From here, we will find λ first.

* The eigenvalue λ tells whether the special vector x is stretched or shrunk or reversed
 $\lambda > 1$ $\lambda < 1$ $\lambda = -1$

Example 1: $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, Find its λ 's and x 's.

$$\text{Now, } \det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 8 = 0$$

$$\Rightarrow \lambda = 2, 4 \rightarrow \lambda_1 = 2, \lambda_2 = 4$$

le.

$$\begin{array}{l|l}
 \text{Now, } A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} & \text{Now } A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\
 \therefore (A - \lambda I)x_1 = 0 & \therefore (A - \lambda I)x_2 = 0 \\
 \rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x_1 = 0 & \rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x_2 = 0 \\
 \rightarrow x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} & \rightarrow x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
 \hookrightarrow \text{eigenvector 1} & \hookrightarrow \text{eigenvector 2}
 \end{array}$$

Example 2: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, find its λ 's and x 's.

Solⁿ: A is a singular matrix here. It has one dependent column.
 So, one of its eigenvalue is 0.

$$\text{Now, } \det(A - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(4-\lambda) - 4 = 0$$

$$\Rightarrow 4 - \lambda - 4\lambda + \lambda^2 - 4 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda = 0$$

$$\Rightarrow \lambda = 5, 0. \quad \therefore \lambda_1 = 5, \lambda_2 = 0$$

$$\begin{array}{l|l}
 \text{Now, for } \lambda = 5, & \text{for } \lambda = 0, \\
 \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 & \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \\
 \rightarrow x_1 = 2, x_2 = 4 & \Rightarrow x_1 = 2, x_2 = -1 \\
 \therefore \begin{bmatrix} 2 \\ 4 \end{bmatrix} \text{ is eigenvector} & \therefore \begin{bmatrix} 2 \\ -1 \end{bmatrix} \text{ is eigenvector.}
 \end{array}$$

⊗ If $Ax = 0x$, meaning $\lambda = 0$ then this eigenvector x is the nullspace of A

Example 3:

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \leadsto \text{Find eigenvalues and eigenvectors.}$$

Solⁿ:

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$$

$$\text{Now, } \det(A - \lambda I) = 0$$

$$\hookrightarrow \begin{vmatrix} .8 - \lambda & .3 \\ .2 & .7 - \lambda \end{vmatrix} = 0$$

$$\hookrightarrow (.7 - \lambda)(.8 - \lambda) - (.3 \times .2) = 0$$

$$\hookrightarrow \left(\frac{7}{10} - \lambda\right)\left(\frac{4}{5} - \lambda\right) - \frac{3}{50} = 0$$

$$\hookrightarrow (25 - 50\lambda)\left(\frac{4}{5} - \lambda\right) - 3 = 0$$

$$\hookrightarrow 28 - 35\lambda - 40\lambda + 50\lambda^2 - 3 = 0$$

$$\hookrightarrow 50\lambda^2 - 75\lambda + 25 = 0$$

$$\hookrightarrow 2\lambda^2 - 3\lambda + 1 = 0 \rightarrow \lambda = 1, \frac{1}{2}$$

Using, $\lambda = 1$

$$\begin{bmatrix} -.2 & .3 \\ .2 & -.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$\hookrightarrow$ This leads to infinite solution.

So taking a suitable basis: $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix}$

Using, $\lambda = 0.5$

$$\begin{bmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

↳ Also infinite solution

So taking a suitable basis: $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

⊗ An eigenvector x of A is also an eigenvector of every A^n . Then $A^n x =$

Proof: $Ax = \lambda x$

$$\Rightarrow A^2 x = A \lambda x$$

$$\Rightarrow A^2 x = \lambda \cdot \lambda x$$

$$\Rightarrow A^2 x = \lambda^2 x$$

$$\vdots$$
$$\Rightarrow A^n x = \lambda^n x$$

This is not really
a proof. Just basic
calculation that you
would do repetitively.

This can be proved
using mathematical induction.

⊗ Permutation matrix, P has eigenvalues 1 and -1 .

Proof:

P is a matrix which has swapped rows of I .

So applying P multiple times eventually returns
to the original ordering. So

↳ $P^k = I$. (Here k is the number of turns)

Now, $Px = \lambda x$

$$\Rightarrow P^2 x = \lambda^2 x$$

λ^u

$$\Rightarrow P^k x = \lambda^k x.$$

$$\Rightarrow Ix = \lambda^k x$$

$$\Rightarrow \lambda^k = I$$

$$\Rightarrow |\lambda^k| = 1 \leadsto \lambda^k - 1 = 0$$

$\hookrightarrow$ Here, the simplest root can be ± 1 .
the roots would be larger and complex
depending on the value of k .

$$\therefore \lambda = \pm 1$$

Eigenvalues of AB and $A+B$

If x is an eigenvector of both A and B ,
with λ and β respectively then:

$$Ax = \lambda x, \quad Bx = \beta x.$$

Now,

$$ABx = A(Bx)$$

$$= A(\beta x) = \beta(Ax) = \beta \lambda x = (\lambda \beta)x$$

Similarly it can be shown that, $BAx = (\lambda \beta)x$

$$\therefore ABx = BAx = (\lambda \beta)x$$

$\hookrightarrow$ So x is an eigenvector of AB and BA

with eigenvalue $\lambda\beta$.

Again,

$$(A+B)x = Ax + Bx = \lambda x + \beta x = (\lambda + \beta)x$$

So same set of eigenvectors x are in AB and $A+B$ as well.

⊗ A and B share the same n independent eigenvectors if and only if $AB = BA$

Determinants, Trace and Eigenvalues

⊗ A triangular matrix (upper or lower) has a simple determinant:

$\det(A)$ = product of diagonal entries.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}$$

$$\hookrightarrow A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ 0 & a_{22} - \lambda & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} - \lambda \end{bmatrix}$$

$$\hookrightarrow \det(A - \lambda I) = (a_{11} - \lambda)(a_{22} - \lambda) \dots (a_{nn} - \lambda) = 0$$

$$\hookrightarrow \left. \begin{array}{l} \lambda_1 = a_{11} \\ \lambda_2 = a_{22} \\ \vdots \\ \lambda_n = a_{nn} \end{array} \right\} \text{⊗ The eigenvalues of a triangular matrix always lie along its diagonal.}$$

⊗ The determinant of a square matrix A is equal to the product of its eigenvalues.

* The determinant of a square matrix A is equal to the product of its eigenvalues: $\det(A) = \lambda_1 \lambda_2 \dots \lambda_n$ 
 $\hookrightarrow$ This will be proved later on.

* The trace of a matrix (sum of diagonal elements) equals the sum of its eigenvalues: $\text{trace}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$ 
 $\hookrightarrow$ Also later on

Both of their proof requires the concept of Diagonalization.

Imaginary Eigenvalues

$\hookrightarrow$ There exists some that will have complex eigenvectors.

* One of the example would be rotation matrix.

$\hookrightarrow \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rightarrow \text{rotates } 90^\circ \rightarrow$ all vectors are rotating, and so direction is changed, so no real eigenvalues.

Assuming, $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

$$\hookrightarrow (A - \lambda I)x = 0$$

$$\hookrightarrow \begin{bmatrix} 0-\lambda & -1 \\ 1 & 0-\lambda \end{bmatrix} x = 0$$

$$\det(A - \lambda I) = 0$$

$$\hookrightarrow \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i$$

* Another example would be non-symmetric matrix.

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$$

$$\hookrightarrow \det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -3 \\ 3 & 1-\lambda \end{vmatrix} = 0$$

$$\hookrightarrow (1-\lambda)^2 + 9 = 0$$

$$\hookrightarrow 1 - 2\lambda + \lambda^2 + 9 = 0$$

$$\hookrightarrow \lambda^2 - 2\lambda + 10 \Rightarrow \lambda = 1 \pm 3i.$$

